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COURSE OUTLINE

UNIT CODE: BIT 324
UNIT TITLE: Mobile Communication Network Design
LECTURE HOURS: 45
PRE-REQUISITES:

1.0 COURSE PURPOSE

To provide students with comprehensive knowledge and practical understanding of mobile communication network design, architecture, technologies, standards, and regulatory frameworks in modern telecommunications systems.

2.0 COURSE LEARNING OUTCOMES

By the end of this course, the student should be able to:

1. Explain the evolution of mobile communication systems from 1G to 4G and beyond.
2. Analyze mobile network architectures including GSM, CDMA, UMTS, and All-IP networks.
3. Evaluate OSS/BSS systems and their role in telecom business operations.
4. Apply multiple access techniques in wireless communication system design.
5. Design basic cellular systems considering frequency reuse, handoff strategies, and capacity improvement.
6. Examine regulatory frameworks, telecom standards, and security mechanisms in mobile networks.
7. Assess mobility management solutions including Mobile IP (MIPv4 and MIPv6).
8. Evaluate integration strategies for Cellular and WLAN networks.

COURSE CONTENT DISTRIBUTION

Week	Sub-Topic	Remarks
1	Introduction to Mobile Communication Network Design – Evolution of Wireless Systems (1G, 2G, 2.5G, 3G, 4G)	
2	Major 2G Standards and 3G Wireless Systems	
3	Wireless LANs (WLANs), Cellular–WLAN Integration, All-IP Networks	
4	OSS/BSS Overview – Architecture, Framework, OSS Components	
5	BSS Components – CRM, Billing, Mediation Systems, Order Management	
6	Telecom Standards and Policies – Regulation, Regulatory Bodies, TMF & eTOM Framework	
7	CAT ONE / Assignment One	
8	Multiple Access Techniques – FDMA, TDMA, CDMA, SDMA	
9	Cellular Concept – Frequency Reuse, Channel Assignment, Handoff, Capacity Improvement	
10	CAT TWO / Assignment Two	
11	GSM Architecture – Signaling Protocols, Identifiers, Channels, Security	
12	CDMA (IS-95), Mobility & Radio Resource Management	
13	Mobile IP (MIPv4 & MIPv6), Mobility Management in Wireless Networks Cellular and WLAN Integration, Heterogeneous Networks, Step Towards 4G	
14	CAT2 and assignment feedback	
15	End of Semester Examination	

3.0 TEACHING METHODOLOGY

- Blended learning (online and face-to-face lectures)
- Case studies
- Group discussions and presentations
- Practical demonstrations and simulations

4.0 INSTRUCTIONAL MATERIALS

- Whiteboard and markers
- LCD projector
- Smartboards
- Learning Management System (LMS)
- Internet-enabled computers
- Telecom simulation software

5.0 COURSE EVALUATION

- CAT 1 = 10%
- CAT 2 = 10%
- Assignments = 10%
- Final Examination = 70%

TOTAL = 100%
Pass Mark = 40%

COURSE CORE TEXTBOOKS

1. Elmasry, G. F. (2012). *Tactical Wireless Communications and Networks: Design Concepts and Challenges*. John Wiley & Sons Ltd.
2. Hanzo, L., Webb, W., & Keller, T. (2000). *Single and Multi-Carrier Quadrature Amplitude Modulation: Principles and Applications for Personal Communications, WLANs and Broadcasting*. John Wiley & Sons Ltd.

REFERENCE TEXTBOOKS

1. Stallings, W. (2014). *Wireless Communications and Networks*. Pearson Education.
2. Rappaport, T. S. (2002). *Wireless Communications: Principles and Practice*. Prentice Hall.

COURSE JOURNALS

1. IEEE Communications Magazine
2. IEEE Transactions on Wireless Communications
3. International Journal of Communication Systems